

Approximating another vector state along a unitary path

MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx

theorem

A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. A representation is a unital complex star homomorphism into the bounded operators on a complex Hilbert space. Irreducibility means nonzero action and no proper nonzero closed reducing subspace. We use the inner product conjugate linear in its first argument and write $\varphi_{\pi,v}(a) = \langle v, \pi(a)v \rangle$ for the vector functional. Let $\pi : A \rightarrow B(H)$ be irreducible with $H \neq 0$, let $\sigma : A \rightarrow B(K)$ be any representation, and choose unit vectors $\xi \in H$ and $\eta \in K$. For every finite $F \subset A$ and $\varepsilon > 0$, there exist a unitary $u \in U(A)$ and a norm-continuous path from 1 to u such that $|\varphi_{\pi,\pi(u)\xi}(a) - \varphi_{\sigma,\eta}(a)| < \varepsilon$ for all $a \in F$.

Assumptions

The spaces H and K are complete complex Hilbert spaces. Both representations are unital; only π must be irreducible. There are no initial stage-state tests and no protected finite set.

Conclusion

The approximating state is obtained from the specified starting vector ξ by a unitary in the identity path component of $U(A)$. The approximation is on the prescribed finite set at the endpoint.

The conclusion supplies a path but makes no near-centrality claim and no uniform approximation claim along that path. It does not identify the two representations or give exact equality of their states on all elements.

Proof route

Keep the given vectors $\xi \in H$ and $\eta \in K$ fixed. Purification supplies a new unit vector $\zeta \in H$ that realizes the target state on a sufficiently large stage m . To move ξ to ζ , use the fixed-stage transport lemma at stage 0, where its tests are automatic. These are two different stages: m controls approximation, whereas stage 0 supplies transport without any near-centrality requirement.

▼ Proof steps

1. Choose m so that each $a \in F$ has an approximant $b_a \in \iota_m(A_m)$ with $\|a - b_a\| < \varepsilon/3$. Apply finite-stage purification to the given π , σ , and η at stage m . Its hypotheses hold because π is nonzero and irreducible, both representations are unital, and $\|\eta\| = 1$. It gives $\zeta \in H$ with $\|\zeta\| = 1$ and $\varphi_{\pi,\zeta}(b) = \varphi_{\sigma,\eta}(b)$ for all $b \in \iota_m(A_m)$. This is a new vector; the prescribed starting vector ξ is not replaced.
2. Use the fixed-stage exact-transport lemma, not the protected-finite-set theorem. For a prescribed stage n and a positive commutator tolerance, it supplies a $\delta > 0$ before the pair of unit vectors is chosen. Its hypothesis compares their vector functionals on the matrix units of that stage. Set $n = 0$ and take the auxiliary commutator tolerance to be 1; its commutator conclusion will not be needed.
3. The inputs to this transport lemma are $\xi, \zeta \in H$ in the same representation π , not ξ and η from different spaces. Since $A_0 = \mathbb{C}$ and $e_{00}^{(0)} = 1$, unitality and the two unit-vector norms give $|\varphi_{\pi,\xi}(e_{00}^{(0)}) - \varphi_{\pi,\zeta}(e_{00}^{(0)})| = |1 - 1| = 0 < \delta$. This is the only matrix-unit test at stage 0. Thus the lemma supplies u and a path p from 1 to u with $\pi(u)\xi = \zeta$. No comparison between these states on the larger stage m , and no norm closeness of ξ and ζ , is required.
4. For each $a \in F$, the two states agree at b_a and are contractive, so $|\varphi_{\pi,\zeta}(a) - \varphi_{\sigma,\eta}(a)| \leq 2\|a - b_a\| < 2\varepsilon/3 < \varepsilon$. Substitute $\zeta = \pi(u)\xi$ to obtain the asserted endpoint estimate.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx
- Finite-stage purification of the target state —
MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage
- Exact transport path, used at stage zero —
MathlibAnnex.CStarAlgebra.CAR.exists_delta_exact_unitary_path_apply_eq_and_stage_commutator
- Representation and nonzero irreducibility conventions —
MathlibAnnex.Analysis.CStarAlgebra.Representation.IsIrreducible
- Equivalence of the two irreducibility interfaces on a nonzero Hilbert space —
MathlibAnnex.Analysis.CStarAlgebra.Representation.isIrreducible_iff_starAlgHom

Lean source signature (exact)

```
theorem exists_unitary_crossRepresentation_path_approx
  {H K : Type*}
  [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
  [Nontrivial H]
  [NormedAddCommGroup K] [InnerProductSpace ℂ K] [CompleteSpace K]
  (rho : Representation Limit H) (hrho : StarAlgHom.IsIrreducible rho)
  (sigma : Representation Limit K) (xi : H) (eta : K)
  (hxi : ‖xi‖ = 1) (heta : ‖eta‖ = 1)
  (F : Finset Limit) {epsilon : ℝ} (hepsilon : 0 < epsilon) :
  ∃ u : unitary Limit, ∃ p : Path 1 u, ∀ a ∈ F,
    ‖Representation.vectorFunctional rho (rho (u : Limit) xi) a -
      Representation.vectorFunctional sigma eta a‖ < epsilon
```

In the source	Mathematical meaning
<code>H; K; rho; hrho; sigma</code>	Complete complex Hilbert spaces H, K , with $H \neq 0$; an irreducible unital representation $\pi : A \rightarrow B(H)$ named rho, and any unital representation $\sigma : A \rightarrow B(K)$.
<code>xi : H; eta : K; hxi; heta</code>	The prescribed vectors $\xi \in H, \eta \in K$ both have norm 1.
<code>F : Finset Limit; hepsilon : 0 < epsilon</code>	The finite endpoint-test set $F \subset A$ and tolerance $\varepsilon > 0$.
<code>∃ u : unitary Limit, ∃ p : Path 1 u</code>	One unitary $u \in U(A)$ joined to 1 by a norm-continuous unitary path p . The path need not nearly centralize F .
<code>Representation.vectorFunctional rho (rho (u : Limit) xi) a</code>	Evaluate the state of the transported vector at a : $\varphi_{\pi, \pi(u)\xi}(a) = \langle \pi(u)\xi, \pi(a)\pi(u)\xi \rangle.$
<code>∀ a ∈ F, ... - Representation.vectorFunctional sigma eta a < epsilon</code>	For every $a \in F$, $ \varphi_{\pi, \pi(u)\xi}(a) - \langle \eta, \sigma(a)\eta \rangle < \varepsilon$. There are no initial stage-state tests and no assertion about intermediate state values along p .

Further source notes: In the linked proof, zeta is the new finite-stage realization of the state of eta; `htrans xi zeta` applies the fixed-stage lemma inside rho at stage zero. The signature imposes no initial comparison of the states of xi and eta.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx
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